

# Uncertainty-Aware Latent Diffusion for Brain Tumour Segmentation with Missing Modalities

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**Abstract.** Brain tumour segmentation from magnetic resonance imaging normally assumes that all four imaging sequences are available, but in clinical practice one or more are frequently missing. Existing methods for incomplete inputs maintain accuracy by learning modality-invariant representations or by distilling from a full-modality teacher, yet they return a single segmentation mask regardless of how much information the input contains. A prediction made from one sequence is therefore presented in the same form as one made from four, and the reader is given no indication of how confident the model is. This paper proposes a three-dimensional conditional latent diffusion framework that produces several plausible segmentations for a given study, together with a voxel-wise map of their disagreement. Two volumetric autoencoders compress images and annotations before diffusion, which makes sampling over  $128^3$  volumes practical, and modality availability is randomly resampled during training so that a single model serves all fifteen acquisition configurations. We further show that averaging predictive variance over the whole volume conflates confidence with lesion size, and propose a restricted average that remains comparable as sequences are withdrawn. Evaluations on 2,601 studies from BraTS 2023 and 2024 indicate that the method maintains whole-tumour accuracy under incomplete inputs, and that the predicted uncertainty increases as sequences are removed.

**Keywords:** Brain Tumour Segmentation · Latent Diffusion Model · Missing Modalities · Uncertainty Estimation · Magnetic Resonance Imaging

## 1 Introduction

Segmenting gliomas on magnetic resonance imaging (MRI) supports diagnosis, surgical planning and the assessment of treatment response. The standard protocol acquires four complementary sequences: T1, contrast-enhanced T1 (T1CE), T2 and FLAIR. No single sequence reveals the whole lesion, since enhancement marks the active core while oedema is clearest on FLAIR. Manual delineation is slow and varies between readers, which has made automated segmentation a central problem in medical image analysis [16].

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Deep encoder–decoder architectures [19, 29, 10] perform well on public benchmarks. A parallel line of work addresses an obstacle these models ignore: the complete protocol is often unavailable, because examinations are shortened, sequences are discarded for motion artefacts, or contrast is contraindicated. Methods for incomplete multimodal segmentation respond by learning representations defined for any available subset [8, 6], by distilling from a full-modality teacher [28], or by separating representation learning from fusion [21].

These methods share an assumption that is rarely stated: that the output should be a single deterministic mask. Their evaluations report how accuracy degrades as sequences are removed, which shows that the prediction depends on the available evidence. The prediction itself, however, is returned in the same form either way. A boundary inferred from FLAIR alone looks no different from one inferred from all four sequences, so the clinician has no indication of how confident the model is.

Generative methods model a distribution over plausible masks rather than one answer [12, 4, 26], and the spread across samples shows what the model finds ambiguous. Validating that spread is harder than producing it, because the quantity it estimates is rarely observable. The usual solution is a dataset annotated several times over, so that predicted variability can be compared with inter-reader disagreement. That disagreement, however, is a fixed property of the data and cannot be changed.

The missing-modality setting removes this limitation. Withholding a sequence is a deliberate change to how much evidence the model receives. Since withholding is a deterministic transformation of the input, it cannot add information about the annotation, so the true posterior is at least as dispersed after removal as before. This gives a behaviour that can be predicted in advance rather than only observed: a calibrated model should report more uncertainty as sequences are withdrawn. Evaluating each case under every subset makes the comparison paired, so lesion difficulty cancels and the available evidence is the only variable that changes.

We present a three-dimensional conditional latent diffusion framework for whole-tumour segmentation from incomplete multi-sequence MRI, and use it to test this behaviour. Our contributions are:

1. **Volumetric latent diffusion with modality dropout.** We extend latent diffusion segmentation to 3D multi-sequence MRI and train it with randomly resampled modality availability, so that one model serves all fifteen acquisition configurations rather than one model per protocol.
2. **Segmentation under every acquisition configuration.** We report whole-tumour accuracy for all fifteen subsets of the four sequences on 2,601 studies from BraTS 2023 and 2024, which shows how far performance can be maintained as sequences are withdrawn.
3. **Uncertainty that is measured and validated under control.** We show that averaging predictive variance over the whole volume mixes confidence with lesion size, and restrict the average to the region where samples can disagree. Because the same case is evaluated under every subset, the resulting

quantity can be tested against the behaviour the data processing inequality prescribes as evidence is removed, giving a paired comparison that datasets with fixed annotation ambiguity cannot provide.

## 2 Related Work

### 2.1 Incomplete Multimodal Tumour Segmentation

Brain tumour segmentation benchmarks assume four co-registered MRI sequences, but clinical acquisitions often deviate. Examinations are shortened, series are corrupted by motion, and contrast administration may be contraindicated. A model that requires the complete protocol is therefore unusable on a large share of real studies, and the literature on incomplete multimodal segmentation addresses this in three main ways.

The first synthesises what is absent, generating the missing sequence and passing the completed set to a conventional segmentation network. This leaves the segmentation architecture unchanged, but accuracy then depends on the quality of the synthesis, and errors introduced there carry through into the mask. The second learns a representation that does not depend on which sequences are present. HeMIS [8] pools per-modality features using statistics that are defined for any subset, so a single model accepts arbitrary inputs. RFNet [6] refines this with region-aware fusion that weights modalities differently for different tumour components, since enhancement and oedema are visible in different sequences. The third distils a full-modality teacher into a student that observes only a subset, transferring knowledge the student cannot obtain from its own input. Transformer-based approaches such as mmFormer [28] combine modality-specific encoders with inter-modality attention to model long-range dependencies across sequences.

A recent refinement separates representation learning from fusion. UniME [21] first pretrains a single shared encoder with masked self-supervision, so that a modality-agnostic representation is established before any fusion mechanism is introduced. Parallel modality-specific encoders are attached only afterwards, and their features are combined at multiple scales. Decoupling the two stages reduces the burden on the fusion module, which no longer has to learn what each sequence contributes and how to combine them at the same time.

These methods differ in mechanism but agree in output: each returns a single deterministic mask, whatever subset it was given. Accuracy is reported as a function of the available sequences, but the confidence attached to the prediction is not. A segmentation produced from one sequence is therefore presented in the same form as one produced from four. How a model’s expressed confidence should respond when evidence is withdrawn is, to our knowledge, not addressed in this literature.

### 2.2 Generative Approaches to Segmentation

Discriminative segmentation networks, from fully convolutional architectures [14] through U-Net [19] and its successors [29, 10], learn a mapping from image to

mask. The mapping is one-to-one by construction: for a given input the model has exactly one answer, so the disagreement observed between expert annotators cannot appear in the output.

Generative formulations relax this. Latent-variable models attach a stochastic code to a segmentation backbone and decode different samples of that code into different masks. The Probabilistic U-Net [12] draws a global latent from a conditional prior, and PHiSeg [4] extends the idea to a hierarchy of latents at several resolutions. Their expressiveness is limited by the size of that latent and by the tendency of the KL term to make the decoder ignore it.

Diffusion models [9] avoid this limit by building the distribution through iterative refinement rather than a single stochastic bottleneck. Conditioning a denoising network on the input image turns generation into segmentation, and repeated sampling gives an ensemble from one set of weights [25, 1]. MedSegDiff [26] strengthens the conditioning with frequency-domain filtering of the guidance signal. The cost is that every sample requires many network evaluations at full resolution, which is acceptable in two dimensions and prohibitive in three.

Latent diffusion [18] removes that cost by moving the process into a compressed space learned by an autoencoder. The approach has been used for medical segmentation by LSegDiff [24] and LDSeg [27]. On the other hand, MedSegLatDiff [22] pairs two autoencoders with a latent diffusion model and shows that a reconstruction loss weighted towards the foreground is needed when targets are small, while LatentFM [23] replaces the diffusion trajectory with flow matching in the same latent space to shorten sampling. Uncertainty arising from incomplete inputs has also begun to receive attention, including flow-matching formulations [17]. These methods operate on two-dimensional images with all inputs present, and evaluate on datasets whose ambiguity is fixed by the annotations. The present work extends this line to volumetric data and to inputs that are deliberately incomplete.

### 3 Method

#### 3.1 Problem Formulation

We formulate brain tumour segmentation from incomplete multi-sequence MRI as conditional generation in a learned latent space. Let

$$\mathcal{D} = \{(X^{(i)}, Y^{(i)})\}_{i=1}^N \quad (1)$$

denote a dataset of multi-sequence brain MRI studies, where  $X^{(i)} \in \mathbb{R}^{H \times W \times D \times M}$  is a co-registered volume with  $M = 4$  sequences (FLAIR, T1ce, T1, T2) and  $Y^{(i)} \in \{0, 1, 2, 4\}^{H \times W \times D}$  is the voxel-wise expert annotation. We target the *whole tumour* (WT), the union of the necrotic core, peritumoural oedema and enhancing tumour, giving the binary mask

$$S = \mathbb{K}[Y > 0] \in \{0, 1\}^{H \times W \times D}. \quad (2)$$

A single binary region is the setting in which probabilistic segmentation is conventionally studied and evaluated [12, 4], and it keeps the uncertainty estimates of Section 3.4 interpretable: dispersion over one foreground class has an unambiguous meaning, whereas dispersion aggregated over overlapping regions of very different size and difficulty does not.

*Modality availability.* In routine practice the four sequences are frequently incomplete: acquisitions are shortened, corrupted by motion, or omitted by protocol. We model availability explicitly with a binary vector  $\mathbf{m} \in \{0, 1\}^M \setminus \{\mathbf{0}\}$ , giving  $2^M - 1 = 15$  distinct configurations, and define the observed volume as

$$X_{\mathbf{m}} = X \odot \mathbf{m}, \quad (3)$$

where  $\odot$  broadcasts  $\mathbf{m}$  across the spatial dimensions so that absent sequences are set to zero. Equation (3) is a deterministic map that can only destroy information, which is the property Section 3.4 relies on.

*Objective.* Rather than learning a point estimate  $f : X_{\mathbf{m}} \mapsto S$ , we learn the conditional distribution  $p_{\theta}(S | X_{\mathbf{m}})$ , from which multiple plausible annotations can be drawn. A deterministic model returns one boundary whatever the input contains, whereas a distribution can widen when the evidence supporting that boundary is removed.

### 3.2 Volumetric Perceptual Compression

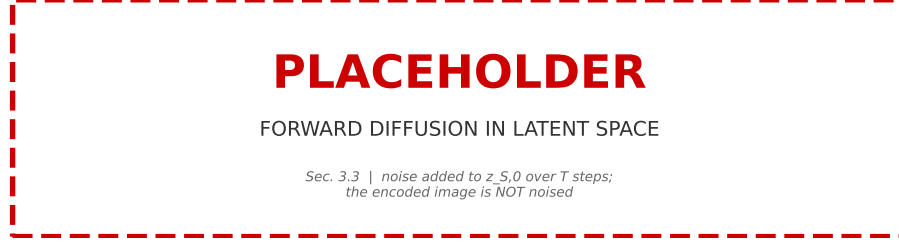
Operating a diffusion model directly on  $128^3$  volumes is prohibitive: the denoising network must be evaluated once per sampling step, and drawing several samples per case multiplies that cost again. We therefore follow the latent diffusion paradigm [18] and move both the image and the annotation into a compressed space before diffusion.

We train two separate variational autoencoders [11]. Their inputs have markedly different statistics. One is continuous and scanner-dependent, the other is piecewise-constant binary structure, and a single shared encoder would have to compromise between them. Each encoder reduces the spatial resolution by a factor of four per axis, from  $128^3$  to  $32^3$ , a  $64\times$  reduction in the number of spatial positions. In two dimensions this compression mainly saves time; in three it is what keeps the model trainable within a fixed memory budget.

*Image autoencoder.* The encoder  $\mathcal{E}_X$  maps the masked volume  $X_{\mathbf{m}}$  to a posterior  $\mathcal{N}(\mu_X, \sigma_X^2)$  over latents, and we take its mean  $z_X = \mu_X \in \mathbb{R}^{32^3 \times C_X}$  as the conditioning signal. Training minimises reconstruction error against the observed sequences together with a Kullback–Leibler term,

$$\mathcal{L}_X = \|X_{\mathbf{m}} - \mathcal{D}_X(z_X)\|_2^2 + \beta_X D_{\text{KL}}(\mathcal{N}(\mu_X, \sigma_X^2) \| \mathcal{N}(0, I)). \quad (4)$$

Modality dropout is applied during autoencoder training as well as during diffusion training, so that the encoder learns to represent partial inputs rather than encountering them for the first time at inference.



**Fig. 1.** Forward diffusion in latent space. Gaussian noise is added to the encoded annotation  $z_{S,0}$  over  $T$  steps until the representation is indistinguishable from  $\mathcal{N}(0, I)$ . The encoded image is not noised; it conditions the reverse process.

*Annotation autoencoder.* The second autoencoder compresses the binary mask  $S$  of Eq. (2). Reconstruction combines a weighted binary cross-entropy with a soft Dice term,

$$\mathcal{L}_S = \underbrace{-\left[w_+ S \log \hat{S} + (1 - S) \log(1 - \hat{S})\right]}_{\text{weighted BCE}} + \underbrace{\left(1 - \frac{2 \text{TP}}{2 \text{TP} + \text{FP} + \text{FN}}\right)}_{\text{soft Dice}} + \beta_S D_{\text{KL}}, \quad (5)$$

where  $\hat{S} = \mathcal{D}_S(z_S)$ , the weight  $w_+ > 1$  upweights tumour voxels, and TP, FP and FN are computed from  $\hat{S}$  and  $S$  without thresholding. Both terms address the same problem from different directions. Tumour occupies a small fraction of each volume, so an unweighted per-voxel objective drives the decoder towards the empty mask, which has low average error but no practical value; the weighting penalises missed tumour voxels more heavily, while the Dice term optimises region overlap directly rather than voxel-wise agreement. The resulting latent  $z_S \in \mathbb{R}^{32^3 \times C_S}$  is the variable the diffusion model generates.

Both autoencoders are frozen after this stage.

### 3.3 Latent Diffusion with Modality Dropout

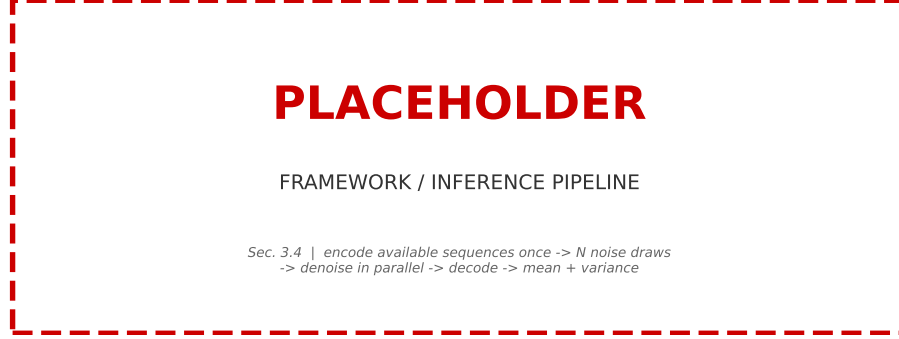
*Forward process.* Diffusion is applied to the annotation latent  $z_{S,0} = \mathcal{E}_S(S)$ . Following DDPM [9], noise is added according to

$$q(z_{S,t} | z_{S,t-1}) = \mathcal{N}(z_{S,t}; \sqrt{\alpha_t} z_{S,t-1}, (1 - \alpha_t)I), \quad (6)$$

which admits the closed form

$$q(z_{S,t} | z_{S,0}) = \mathcal{N}(z_{S,t}; \sqrt{\gamma_t} z_{S,0}, (1 - \gamma_t)I), \quad \gamma_t = \prod_{i=1}^t \alpha_i, \quad (7)$$

allowing any timestep to be sampled directly during training. The process is illustrated in Fig. 1.



**Fig. 2.** Inference. The available sequences are encoded once;  $N$  independent noise draws are denoised in parallel and decoded, yielding  $N$  plausible annotations. Their mean is the prediction and their per-voxel variance is the uncertainty estimate.

*Conditioning.* The denoising network is conditioned on the encoded image by channel-wise concatenation in the latent space,

$$z_{S,t}^{\text{cond}} = z_{S,t} \oplus \pi(z_X), \quad (8)$$

where  $\pi$  is a learned  $1 \times 1 \times 1$  projection that reduces the image latent to a smaller channel count before concatenation, keeping the conditioning from dominating the input by sheer width. Because the two latents share a spatial resolution of  $32^3$ , conditioning requires no resampling or cross-attention.

*Modality dropout.* The availability vector  $\mathbf{m}$  is resampled independently for every training example, uniformly over the 15 non-empty configurations. A single set of parameters therefore covers every acquisition protocol, in contrast to training a separate model per configuration or imputing absent sequences before segmentation. Uniform sampling deliberately over-represents degraded inputs relative to their clinical frequency, so that the sparse configurations, where uncertainty matters most, are not treated as rare corner cases.

*Training objective.* With  $\epsilon \sim \mathcal{N}(0, I)$  and  $t$  drawn uniformly over  $\{1, \dots, T\}$ , the network  $\epsilon_\theta$  is trained to predict the noise added at step  $t$ :

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{z_{S,0}, \mathbf{m}, \epsilon, t} \left[ \|\epsilon - \epsilon_\theta(z_{S,t}^{\text{cond}}, t)\|_2^2 \right]. \quad (9)$$

The expectation over  $\mathbf{m}$  in Eq. (9) is what distinguishes this objective from a standard conditional diffusion loss: the model is optimised over the whole family of degraded inputs rather than a single fixed one.

### 3.4 Sampling and Predictive Uncertainty

At inference, the available sequences are encoded once and  $N$  independent initial noise tensors  $z_{S,T}^1, \dots, z_{S,T}^N \sim \mathcal{N}(0, I)$  are denoised with DDIM [20] under the

same conditioning. Decoding each trajectory gives  $N$  annotations  $S^i = \mathcal{D}_S(z_{S,0}^i)$  for  $i = 1, \dots, N$ , which are samples from the learned posterior  $p_\theta(S \mid X_{\mathbf{m}})$ . Their per-voxel mean and variance,

$$\mu(v) = \frac{1}{N} \sum_{i=1}^N S^i(v), \quad \sigma^2(v) = \frac{1}{N} \sum_{i=1}^N (S^i(v) - \mu(v))^2, \quad (10)$$

give the prediction and the uncertainty map respectively; the final segmentation is  $\mathbb{K}[\mu(v) > \tau]$  with  $\tau = 0.5$ .

The dispersion in Eq. (10) is not noise added to produce variability. The model is trained to sample from  $p(S \mid X_{\mathbf{m}})$ , so the spread across draws is the posterior the model has learned. This differs from Monte Carlo dropout [7], which perturbs the network itself and reports the resulting instability, and from deep ensembles [13], which capture parameter uncertainty but require training several models.

*Restricting the average.* Summarising an uncertainty map by one number requires care. The mean of  $\sigma^2$  over the whole volume is dominated by background: the great majority of voxels are unambiguously healthy tissue, every sample agrees, and their variance is identically zero. Such an average therefore scales inversely with lesion size rather than with confidence, and it is not comparable across modality configurations, because the predicted lesion extent itself depends on  $\mathbf{m}$ . We instead average only over the region in which the samples are able to disagree: the set  $\Omega$  of voxels that are annotated tumour, or that at least one sample labels tumour. Writing  $|\Omega|$  for its size,

$$U(X_{\mathbf{m}}) = \frac{1}{|\Omega|} \sum_{v \in \Omega} \sigma^2(v). \quad (11)$$

Equation (11) measures how much plausible annotations disagree in the region where they can disagree, and is the only form of the quantity that stays comparable across the configurations we compare below.

*Expected behaviour under information loss.* The masking in Eq. (3) is a deterministic function of  $X$ , so the data processing inequality [5] gives, for any nested pair of configurations  $\mathbf{m}' \subseteq \mathbf{m}$ ,

$$I(S; X_{\mathbf{m}'}) \leq I(S; X_{\mathbf{m}}), \quad \text{hence} \quad H(S \mid X_{\mathbf{m}'}) \geq H(S \mid X_{\mathbf{m}}). \quad (12)$$

Removing a sequence cannot create information about the annotation, so the true posterior is at least as dispersed after removal as before, and a well-calibrated model should report uncertainty that follows the same ordering. Two qualifications apply. The inequality holds only for *nested* configurations, so whether T1CE is more informative than T1 is an empirical question rather than a consequence of the bound. It also constrains the true posterior and not the model's estimate of it, so a poorly calibrated model may stay confident while the supporting evidence disappears.

Since every case is evaluated under all 15 configurations, this effect can be measured within each case. For case  $i$  we report

$$\Delta U_i(\mathbf{m}) = U(X_{\mathbf{m}}^{(i)}) - U(X_{\mathbf{1}}^{(i)}), \quad (13)$$

where  $\mathbf{1}$  denotes the fully observed input. Lesion size and case difficulty are shared between the two terms and cancel, leaving the contribution of the removed sequences.

## 4 Experiments

### 4.1 Dataset and Preprocessing

We evaluate on the BraTS 2023 and 2024 adult glioma collections [16, 3, 2], which provide 1,251 and 1,350 studies with publicly released annotations, giving 2,601 in total. Each study provides four co-registered, skull-stripped sequences (T1, T1CE, T2 and FLAIR) at  $240 \times 240 \times 155$  with a voxel-wise expert annotation. As described in Section 3.1, we take the whole tumour as the binary target.

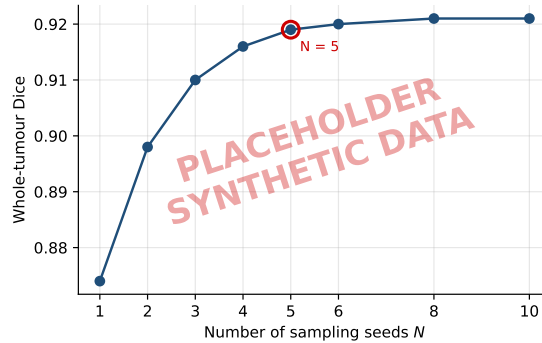
Intensities are normalised per sequence and per study by  $z$ -scoring over brain voxels only, so that background air does not bias the statistics and scanner-dependent ranges are placed on a common scale. Each study is then cropped to a fixed  $128^3$  window centred on the centroid of the annotation bounding box. A fixed crop allows annotation latents to be computed once and cached, and this window retains the complete tumour for 97% of studies.

Studies are partitioned patient-wise into training, validation and test sets in a 70:20:10 ratio with a fixed seed, giving 1,821, 520 and 260 studies respectively. All configurations of a given study fall in the same partition, so no case is ever evaluated under one subset of sequences while having been trained on under another.

### 4.2 Implementation Details

*Autoencoders.* Both autoencoders use a residual convolutional encoder-decoder with two residual units per level and two downsampling stages, mapping  $128^3$  inputs to  $32^3$  latents. The image autoencoder uses channel widths (64, 128, 256) and a KL weight of  $\beta_X = 10^{-4}$ ; the annotation autoencoder uses (32, 64, 128), a latent width of  $C_S = 4$  and  $\beta_S = 10^{-2}$ . The larger KL weight on the annotation branch reflects its simpler target, since a binary mask needs less latent capacity than a four-sequence volume.

Both are trained for 200 epochs with AdamW [15], learning rate  $10^{-4}$ , weight decay  $10^{-4}$  and batch size 32, with early stopping after 25 validation checks without improvement. Modality dropout is applied when training the image autoencoder as well as the diffusion model, so the encoder learns to represent partial inputs rather than meeting them for the first time at inference. Both autoencoders are frozen thereafter.



**Fig. 3.** Whole-tumour Dice against the number of sampling seeds  $N$ , with all four sequences available. The curve flattens after five samples, which is the value used throughout.

*Diffusion model.* The denoising network is a 3D U-Net with base width 64, channel multipliers (1, 2, 4) and two residual blocks per level, totalling 31.3M parameters. The image latent is projected from 256 to 64 channels before being concatenated with the noisy annotation latent, so that the conditioning does not dominate the input by width. Training uses  $T = 1000$  timesteps with a linear noise schedule,  $2 \times 10^5$  optimisation steps, batch size 64 and AdamW with learning rate  $10^{-4}$  and weight decay  $10^{-4}$ . The modality vector  $\mathbf{m}$  is resampled independently for every example, uniformly over the 15 non-empty configurations.

*Inference.* Sampling uses DDIM [20] with 50 steps. We draw  $N = 5$  samples per case, average them to obtain the prediction, and threshold at  $\tau = 0.5$ . Figure 3 shows how whole-tumour Dice varies with  $N$  on the full-modality configuration. Accuracy rises steeply for the first few samples and then flattens, so  $N = 5$  sits at the point where further samples cost inference time without improving the ensemble.

All experiments use a fixed random seed and are implemented in PyTorch; training was performed on NVIDIA H200 GPUs.

### 4.3 Evaluation Protocol and Metrics

Every test case is evaluated under all 15 modality configurations, giving a complete case  $\times$  configuration grid. Segmentation quality is reported with the Dice coefficient and Intersection over Union between the thresholded ensemble and the annotation. Predictive uncertainty is reported with  $U(X_{\mathbf{m}})$  from Eq. (11), the mean sample variance restricted to the region where samples can disagree, and with the paired difference  $\Delta U_i(\mathbf{m})$  of Eq. (13) relative to the fully observed input for the same case.

**Table 1.** Reconstruction fidelity of the frozen autoencoders. The annotation autoencoder bounds the segmentation performance attainable in latent space.

| Autoencoder                             | Dice | IoU | SSIM | PSNR |
|-----------------------------------------|------|-----|------|------|
| Image ( $128^3 \rightarrow 32^3$ )      | –    | –   | –    | –    |
| Annotation ( $128^3 \rightarrow 32^3$ ) | –    | –   | –    | –    |

**Table 2.** Segmentation accuracy and predictive uncertainty across all 15 modality configurations, grouped by the number of available sequences.  $\Delta U$  is the paired increase in uncertainty relative to the fully observed input for the same case.

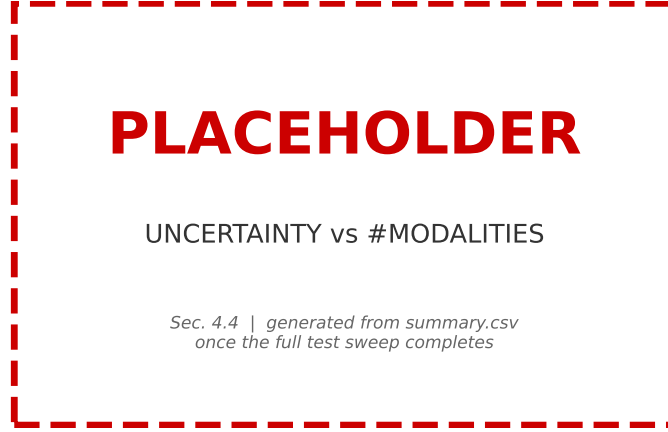
| $ \mathbf{m} $ | Available sequences    | Dice | IoU | $U$ | $\Delta U$ |
|----------------|------------------------|------|-----|-----|------------|
| 4              | FLAIR + T1ce + T1 + T2 | –    | –   | –   | –          |
| 3              | FLAIR + T1ce + T1      | –    | –   | –   | –          |
|                | FLAIR + T1ce + T2      | –    | –   | –   | –          |
|                | FLAIR + T1 + T2        | –    | –   | –   | –          |
|                | T1ce + T1 + T2         | –    | –   | –   | –          |
| 2              | FLAIR + T1ce           | –    | –   | –   | –          |
|                | FLAIR + T1             | –    | –   | –   | –          |
|                | FLAIR + T2             | –    | –   | –   | –          |
|                | T1ce + T1              | –    | –   | –   | –          |
|                | T1ce + T2              | –    | –   | –   | –          |
|                | T1 + T2                | –    | –   | –   | –          |
| 1              | FLAIR                  | –    | –   | –   | –          |
|                | T1ce                   | –    | –   | –   | –          |
|                | T1                     | –    | –   | –   | –          |
|                | T2                     | –    | –   | –   | –          |

Because the autoencoder is frozen, its reconstruction fidelity upper-bounds what any diffusion model operating in that latent space can achieve. We therefore report autoencoder reconstruction separately in Table 1, so that the segmentation results in Table 2 can be read against that ceiling.

#### 4.4 Results

### 5 Conclusion

We presented a three-dimensional conditional latent diffusion framework for whole-tumour segmentation from incomplete multi-sequence MRI. Two volumetric autoencoders compress images and annotations by a factor of four per axis, which makes diffusion over  $128^3$  volumes tractable, and modality dropout during training lets a single model serve all fifteen acquisition configurations. Repeated sampling yields both a consensus segmentation and a voxel-wise map of disagreement between plausible annotations.



**Fig. 4.** Predictive uncertainty against the number of available sequences. Each case contributes one observation per configuration, so the comparison is paired.

Beyond accuracy, we treated modality availability as an experimental variable rather than a nuisance. Because withholding a sequence cannot add information about the annotation, a calibrated model should report more uncertainty as sequences are removed, and evaluating every case under every subset makes that prediction testable within each case. Our results indicate that the predicted uncertainty follows this behaviour, which suggests it reflects genuine information loss rather than sampling noise.

### 5.1 Limitations and Future Work

*Compression cost on small structures.* The autoencoders apply a fixed  $4\times$  downsampling per axis, chosen so that diffusion over whole volumes fits within memory. This is not neutral with respect to structure size. Reconstructing the annotation through the frozen autoencoder recovers the whole tumour almost exactly, but is markedly less faithful for the enhancing core, which is smaller and thinner. Since the diffusion model operates entirely inside this latent space, that gap is a ceiling it cannot exceed, and it is the main reason we report the whole tumour rather than the full BraTS region hierarchy. Adaptive or anisotropic downsampling, or a latent with higher spatial resolution for the annotation branch, would target this directly.

*Region coverage.* We segment the whole tumour only. Extending to tumour core and enhancing tumour is the natural next step, and the analysis above suggests it depends more on the compression scheme than on the diffusion model itself. Confirming that would require repeating the autoencoder study at several downsampling factors and measuring where the reconstruction ceiling for each region falls.



**Fig. 5.** Samples for one case under decreasing modality availability. Rows are configurations ordered from most to least informative; columns are independent sampling seeds, followed by the ensemble and the uncertainty map.

*Reconstruction objective.* The annotation autoencoder combines weighted cross-entropy with a soft Dice term. Losses designed specifically for thin or sparse structures, such as boundary-aware or focal variants, may preserve small regions better at the same compression factor, and we have not compared them.

*Uncertainty baselines.* We compare the predicted uncertainty against the behaviour information theory prescribes, but not against other estimators. Monte Carlo dropout and deep ensembles would place our estimates on a common footing with established methods, and the Probabilistic U-Net would provide a latent-variable comparison. Such a study would also allow calibration metrics that require a reference distribution, which a single fused annotation per case does not support.

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